



Catalyst-free synthesis of 3-hydroxy-3-(alkyl/aryl)indolin-2-ones by addition of organoaluminum reagents to isatins



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ABSTRACT

An efficient synthesis of 3-hydroxy-3-(alkyl/aryl)indol-2-one derivatives has been described by the reactions of isatin with organoaluminum reagents. The reaction is very rapid and yields are high. The protocol is applicable for substituted isatins as well as a variety of organoaluminums.

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The 3-hydroxyoxindole moiety is found in many biologically active molecules such as convolutamydines.¹ This is important structural motif in proteasome inhibitors like TMC-95A/B.² The 3-hydroxyoxindole moiety is also found in various alkaloids such as donaxaridine, dioxibrassinine, welwitindolinone C, and 3-hydroxyglucoisatin.³ The carbon–carbon bond formation has been one of the most studied reactions in the past 10 years.⁴ In these studies, the additions of organozinc to aldehydes are one of the most reliable processes and thus gain much attention from chemists.^{5–9} Among alkylation reagents, trialkylaluminum reagents are more interesting since they are inexpensive and commercially available.¹⁰ Among the organometallic species, organoaluminum reagents stand out for practical applications. The additional advantages of organoaluminum compounds include low toxicities and considerable stabilities.¹¹ The first example of the asymmetric addition of AlEt₃ to aldehydes, catalyzed by a Ti(IV)–BINOL complex, was developed by Chan et al. in 1997.^{12d} Other reports include the titanium catalyzed addition of Me₃Al,^{12b,c} Et₃Al,^{12b} (allyl)AlEt₂,^{12b} and Ph₃Al^{12a} to aldehydes. In addition to this some of the methods have been reported using nickel catalyzed Me₃Al¹¹ or (Me₃Al)₂–DABCO^{11c} to aldehydes (Fig. 1).

Owing to the significance of this organoaluminum reagent motif, the development of these reactions is highly valuable, and the alkylation/arylation of isatins would be the most straightforward approach to this end. Catalyst free reactions have received special attention in recent years.¹³ There are two methods for the synthesis of 3-hydroxy-3-alkyl/aryl-indolin-2-ones which involves using

Grignard reagents (RMgX)¹⁴ or arylboronic acids (RB(OH)₂).¹⁵ Following our recent interest in the synthesis of new 3-hydroxy-

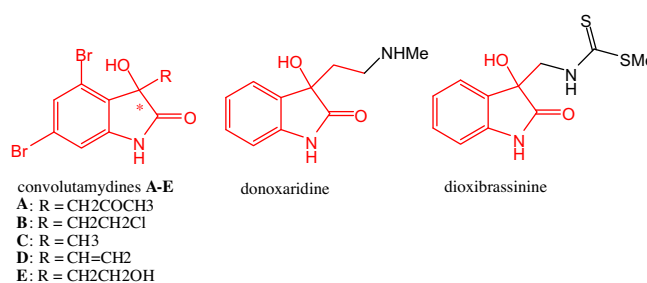
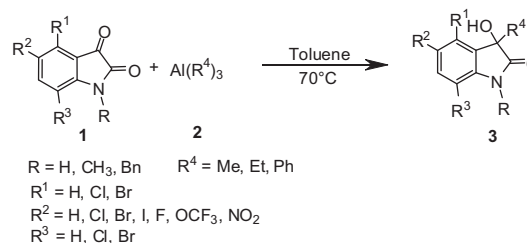


Figure 1. Representative examples of biologically active 3-hydroxy-indolin-2-ones.



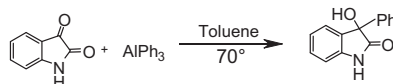
Scheme 1. Synthesis of substituted 3-hydroxy-indolin-2-ones under catalyst-free condition.

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Table 1

Screened solvents for the reaction of isatin with organoaluminum reagents



Entry	Solvent	Condition ^a	Yield ^b (%)
1	DMSO	120 °C, 8 h	55
2	Toluene	rt, 24 h	20
3	Xylene	70 °C, 12 h	55
4	Toluene	70 °C, 3 h	90
5	Dioxane	100 °C, 12 h	40
6	Acetonitrile	Reflux, 12 h	20
7	THF	Reflux, 12 h	20
8	DCM	Reflux, 12 h	20

^a Reaction conditions: isatin (1 equiv) and organoaluminum (2 equiv) in solvent stirred at stipulated time.^b Isolated yields.

3-substituted oxindole,¹⁶ herein, we wish to report the catalyst free efficient synthesis of 3-hydroxy-3-(alkyl/aryl)indolin-2-one derivatives by the organometallic reaction of isatins with trialkyl/aryl aluminum in excellent yields. To the best of our knowledge, however, there have been no examples reported to date that employ this strategy (Scheme 1).

We have initiated our investigation by examining the reaction between free isatin **1** and triphenylaluminum **2** as a model. Initially we have observed that the formation of desired product in less yield when triphenylaluminum(1 mmol) reacted with free isatin(1 mmol) in toluene at rt. Next we focused to examine the reaction in detail to increase the reaction rate and yield. Surprisingly, the extreme increase in the reaction rate and yield (90%) of **3a** was observed when the temperature raised to 70 °C without using any catalyst. And we also observed that there is no change in yield even if the temperature is raised to 100 °C. The reaction completed smoothly within 3 h at 70 °C to give 3-hydroxy-3-(phenyl)indolin-2-one **3a** which was confirmed by analyzing spectral data (Table 2, entry 1). Though the reaction proceeds with 1:1 ratio of reactant, the best results were obtained by 2 equiv ratio of AlPh₃.

Encouraged by these results, next we have studied the model reaction in different solvents such as DMSO, toluene, xylene, dioxane, acetonitrile, THF, and DCM. We have noticed that the reaction proceeded in all the solvents with a different degree of conversion (Table 1). However, toluene was the solvent of choice in terms of reaction time and yield.

The scope of this catalyst-free protocol was investigated under optimal condition¹⁷ and the results are summarized in Table 2. Isatins bearing different substituents on the nitrogen atom and/

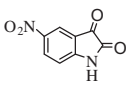
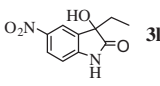
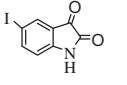
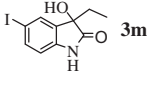
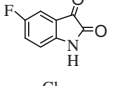
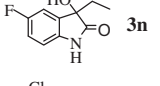
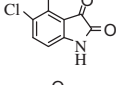
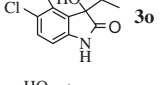
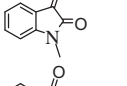
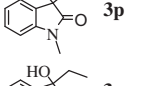
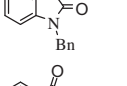
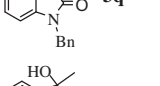
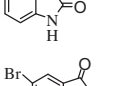
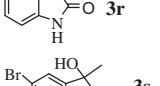
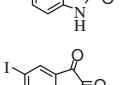
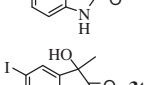
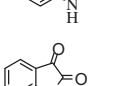
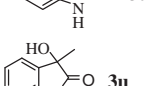
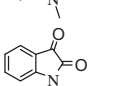
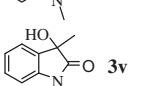
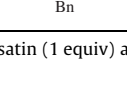
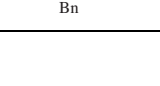
Table 2

3-Hydroxy-3-(alkyl/aryl)-indolin-2-one derivatives formation by using organoaluminum

Entry	Isatin	Al (Alkyl/Aryl) ₃	Product ^a	Time (h)	Yield ^b (%)
1		AlPh ₃	3a	3	90
2		AlPh ₃	3b	3	93
3		AlPh ₃	3c	3	92
4		AlPh ₃	3d	3	93
5		AlPh ₃	3e	3	95
6		AlPh ₃	3f	3	94
7		AlPh ₃	3g	3.5	80
8		AlEt ₃	3h	3.5	85
9		AlEt ₃	3i	3.5	84
10		AlEt ₃	3j	3.5	75
11		AlEt ₃	3k	3.5	80

(continued on next page)

Table 2 (continued)

Entry	Isatin	Al (Alkyl/Aryl) ₃	Product ^a	Time (h)	Yield ^b (%)
12		AlEt ₃	 3l	3.5	78
13		AlEt ₃	 3m	3.5	78
14		AlEt ₃	 3n	3.5	80
15		AlEt ₃	 3o	3.5	78
16		AlEt ₃	 3p	3.5	76
17		AlEt ₃	 3q	3	76
18		AlMe ₃	 3r	3	90
19		AlMe ₃	 3s	3	89
20		AlMe ₃	 3t	3	89
21		AlMe ₃	 3u	3	88
22		AlMe ₃	 3v	3	88

^a Reaction conditions: isatin (1 equiv) and organoaluminum (2 equiv) at 70 °C.^b Isolated yields.

or the aromatic ring could also be alkylated/arylated effectively under this condition. The addition of AlPh₃ **2** to various isatins **1** proceeded readily and resulted into expected product in high yield. Further, we extended this protocol to different substituted isatins. For example, 5-halo isatins reacted with AlPh₃ in the optimized condition and resulted into high yields of the expected product (Table 2, entries 2–4). Other 5-substituted, 7-substituted, and 4,7-substituted isatins also afforded comparatively good yields of products (Table 2, entries 5–7). Then we extended this protocol for the AlMe₃/AlEt₃ addition to different substituted isatins which gave good yield of the products. For example, simple isatin reacted with AlMe₃/AlEt₃ to afford high yield of desired products (Table 2, entries 8 and 18). In addition to this N-substituted isatins (Table 2, entries 16 and 17) were subjected to the AlMe₃/AlEt₃ reaction in an identical condition which resulted in the expected product in less yield. Similarly, 4-substituted, 5-substituted and also 5,7-disubstituted isatins (Table 2, entries 11–13 and 15) participated in the reaction and gave comparatively less yields. During the experiment we observed the reactivity profile of alkyl/aryl Al is as follows AlPh₃ > AlMe₃ > AlEt₃.

In summary, we have developed a novel method for the alkylation/arylation of isatin at 3 position, using tri alkyl/aryl aluminum. This method is applicable for substituted isatins and yields are high. Moreover the present protocol is also amenable for the

synthesis of new novel compounds which are not prepared earlier (Table 2, entries 6–15 and 20). A wide variety of isatins having different substituents were examined. Isatins bearing either an electron-donating or an electron-withdrawing substituent at 4-, 5-, and 7-position gave excellent yield of products. More importantly, aluminum reagents with different aryl nucleophiles can be easily prepared for additions to isatins, furnishing tertiary alcohols. The present method provides direct access for the synthesis of a large number of 3-alkyl/aryl-3-hydroxy indole-2-ones molecules which may be useful in search of lead molecules.

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17. *General procedure*: To a solution of isatin **1** (1 mmol) in toluene (5 mL) was added the solution of trialkylaryl Al **2** (2 mmol, 2 M in toluene) dropwise at RT. Then the mixture was heated at 70 °C and stirred for additional 3–3.5 h (see Table 2). The reaction was monitored by TLC. After completion of the reaction, toluene was evaporated and extracted with ethyl acetate (3 × 5 mL). The combined organic layer was washed with brine solution, dried over anhydrous Na₂SO₄, and concentrated under reduced pressure (rotary evaporator). The crude products were purified by silica gel column chromatography using ethyl acetate/hexanes (4:1–1:1). All compounds were characterized by (MP, NMR, Mass, and IR) spectral data.¹⁸
18. *Spectral data for new compounds*:
 4-Bromo-3-ethyl-3-hydroxyindolin-2-one (Table 2, **3f**): White solid; Mp 160–162 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 3.68 (s, 1H), 6.98 (d, *J* = 8.98 Hz, 1H), 7.26 (d, *J* = 8.98 Hz, 1H), 7.33–7.42 (m, 5H), 8.02 (s, 1H) ppm; ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 93.3, 109.5, 125.8, 126.1, 126.4, 127.1, 129.3, 129.6, 136.4, 139.2, 143.0, 163.0 ppm; IR (KBr): ν = 3374, 3195, 1718, 1611, 1467, 1422, 1303, 1160, 1063, 810, 768, 700 cm^{−1}; MS (EI): *m/z* = 295 (M⁺1).
 4-Bromo-3-ethyl-3-hydroxyindolin-2-one (Table 2, **3i**): White solid; Mp 162–164 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 0.59 (t, *J* = 7.32 Hz, 3H), 1.83 (q, *J* = 7.32 Hz, 2H), 5.50 (s, 1H), 6.79 (dd, *J* = 1.97, 5.93 Hz, 1H), 7.00–7.04 (m, 2H), 10.12 (s, 1H) ppm; ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 4.3, 31.2, 82.1, 113.8, 123.3, 127.3, 128.4, 130.6, 141.8, 177.8 ppm; IR (KBr): ν = 3246, 3177, 1706, 1619, 1472, 1440, 1182, 1082, 930, 826, 730 cm^{−1}; MS (EI): *m/z* = 279 (M⁺Na).
 3-Ethyl-3-hydroxy-5-(trifluoromethoxy)indolin-2-one (Table 2, **3j**): White solid; Mp 158–160 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 0.74 (t, *J* = 7.66 Hz, 3H), 1.92 (q, *J* = 7.66 Hz, 2H), 5.79 (s, 1H), 6.86 (d, *J* = 8.20 Hz, 1H), 7.06 (d, *J* = 8.20 Hz, 1H), 7.17 (s, 1H), 10.06 (s, 1H) ppm; ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 4.3, 31.9, 86.2, 111.1, 111.8, 112.4, 121, 131.8, 133, 156.3, 177.8 ppm; IR (KBr): ν = 3346, 3157, 1709, 1629, 1462, 1240, 1182, 1082, 920, 816 cm^{−1}; MS (EI): *m/z* = 262 (M⁺1).
 5-Chloro-3-ethyl-3-hydroxyindolin-2-one (Table 2, **3k**): White solid; Mp 204–206 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 0.72 (t, *J* = 7.32 Hz, 3H), 1.84 (q, *J* = 7.32 Hz, 2H), 5.72 (s, 1H), 6.78 (t, *J* = 8.54 Hz, 1H), 7.12 (d, *J* = 8.54 Hz, 1H), 7.19 (s, 1H), 10.10 (s, 1H) ppm; ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 4.3, 31.9, 85.4, 110.9, 125.9, 126.3, 126.6, 132.2, 139.0, 177.8 ppm; IR (KBr): ν = 3256, 2942, 1703, 1605, 1490, 1423, 1325, 1185, 1070, 985, 924, 798 cm^{−1}; MS (ESI): *m/z* = 234 (M⁺Na).
 3-Ethyl-3-hydroxy-5-nitroindolin-2-one (Table 2, **3l**): White solid; Mp 209–211 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 0.76 (d, *J* = 7.00 Hz, 3H), 1.92 (q, *J* = 7.00 Hz, 2H), 5.92 (s, 1H), 6.95 (d, *J* = 8.0 Hz, 1H), 8.09–8.14 (m, 2H), 10.6 (s, 1H); ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 4.3, 31.9, 84.9, 108.9, 121.4, 124.7, 131.7, 143.6, 147, 177.8; IR (KBr): ν = 3285, 2927, 2856, 1714, 1624, 1524, 1338, 1177, 1089, 840, 734 cm^{−1}; MS (EI): *m/z* = 223 (M⁺1).
 3-Ethyl-3-hydroxy-5-iodoindolin-2-one (Table 2, **3m**): White solid; Mp 220–227 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 0.71 (t, *J* = 7.62 Hz, 3H), 1.82 (t, *J* = 7.62 Hz, 2H), 5.74 (s, 1H), 6.62 (d, *J* = 8.71 Hz, 1H), 7.45 (d, *J* = 7.62 Hz, 1H), 7.49 (s, 1H), 10.14 (s, 1H) ppm; ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 4.3, 31.9, 85, 90.0, 123.3, 132.4, 135.1, 139.2, 139.8, 177.8 ppm; IR (KBr): ν = 3258, 3184, 2920, 1701, 1626, 1489, 1157, 927, 817, 733 cm^{−1}; MS (EI): *m/z* = 326 (M⁺Na).
 3-Ethyl-5-fluoro-3-hydroxyindolin-2-one (Table 2, **3n**): White solid; Mp 158–160 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 0.72 (t, *J* = 7.55 Hz, 3H), 1.90 (q, *J* = 7.55 Hz, 2H), 5.86 (s, 1H), 6.81 (d, *J* = 8.30 Hz, 1H), 6.86–6.93 (m, 1H), 7.02 (dd, *J* = 7.93, 8.30 Hz, 1H), 10.06 (s, 1H) ppm; ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 6.7, 30.3, 84.4, 110.7, 111.0, 114.1, 132.5, 136.9, 159.5, 179.5 ppm; IR (KBr): ν = 3267, 2933, 1698, 1610, 1485, 1435, 1343, 1175, 1095, 993, 805, 704 cm^{−1}; MS (EI): *m/z* = 218 (M⁺Na).
 4,5-Dichloro-3-ethyl-3-hydroxyindolin-2-one (Table 2, **3o**): Yellow solid; Mp 228–230 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 0.59 (t, *J* = 7.00 Hz, 3H), 1.90 (q, *J* = 7.00 Hz, 2H), 5.79 (s, 1H), 6.73 (d, *J* = 8.00 Hz, 1H), 7.29 (d, *J* = 8.00 Hz, 1H), 10.10 (s, 1H) ppm; ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 4.3, 31.4, 80.3, 109.0, 125.8, 129.1, 129.1, 132.5, 139.1, 177.8 ppm; IR (KBr): ν = 3256, 2942, 1703, 1605, 1490, 1423, 1325, 1185, 1070, 985, 924, 798 cm^{−1}; MS (EI): *m/z* = 247 (M⁺1).
 3-Hydroxy-5-iodo-3-methylindolin-2-one (Table 2, **3t**): Yellow solid; Mp 248–250 °C; ¹H NMR (300 MHz, CDCl₃ + DMSO-*d*₆): δ 1.48 (s, 3H), 5.82 (s, 1H), 6.67 (d, *J* = 8.60 Hz, 1H), 7.48 (s, 1H), 7.58 (s, 1H), 10.03 (s, 1H) ppm; ¹³C NMR (75 MHz, CDCl₃ + DMSO-*d*₆): δ 23.2, 77.4, 90.4, 123.7, 129.5, 136.7, 140.7, 141.3, 177.5 ppm; IR (KBr): ν = 3258, 3184, 2920, 1701, 1626, 1489, 1157, 927, 817, 733 cm^{−1}; MS (ESI): *m/z* = 312 (M⁺Na).